

1 Solution — GIAM 1.4

Problem 10

1.1 Problem

Suppose that 340 pounds of sand must be placed into bags having a 50-pound capacity. Write an expression using either floor or ceiling notation for the number of bags required.

1.2 The Answer

$$\text{number of bags} = \left\lceil \frac{340}{50} \right\rceil = \lceil 6.8 \rceil = 7.$$

Seven bags are needed: six filled to capacity hold $6 \times 50 = 300$ pounds, and a seventh holds the remaining 40 pounds.

1.3 Why Ceiling, Not Floor

The exact quotient is $340/50 = 6.8$, which is *not* a whole number — and you cannot use a fraction of a bag. The two roundings give different counts:

- $\lfloor 6.8 \rfloor = 6$ — the *floor*, counting only the **completely full** bags. Six bags hold just 300 pounds, leaving 40 pounds unbagged. **Too few.**
- $\lceil 6.8 \rceil = 7$ — the *ceiling*, rounding **up** to include the partially-filled seventh bag. **Correct.**

Because any leftover sand still requires a whole additional bag, we must round *up* — and rounding up is exactly what the ceiling function does:

$$\lceil x \rceil = \text{the least integer} \geq x.$$

1.4 The Picture

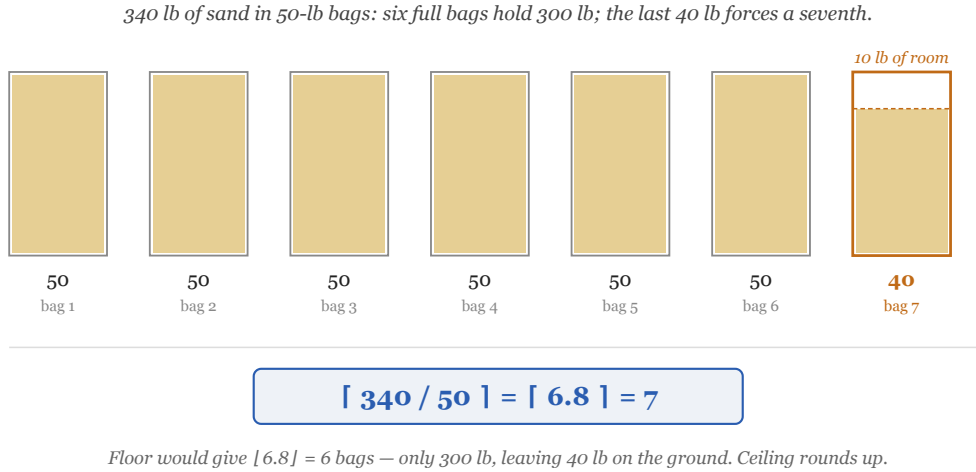


Figure 1.1: Six bags filled to the 50-lb line (300 lb total) plus a seventh, only 80% full, holding the last 40 lb. The expression $\lceil 340/50 \rceil = \lceil 6.8 \rceil = 7$ counts all seven; $\lfloor 6.8 \rfloor = 6$ would leave 40 lb on the ground.

1.5 A Floor Version, If Required

The problem allows *either* floor or ceiling. A ceiling can always be rewritten as a floor using the integer identity

$$\left\lceil \frac{a}{b} \right\rceil = \left\lfloor \frac{a + b - 1}{b} \right\rfloor \quad (a, b \text{ positive integers}).$$

With $a = 340$, $b = 50$:

$$\left\lfloor \frac{340 + 50 - 1}{50} \right\rfloor = \left\lfloor \frac{389}{50} \right\rfloor = \lfloor 7.78 \rfloor = 7.$$

So $\lfloor 389/50 \rfloor$ is an equally valid answer. The “ $+ (b - 1)$ ” nudge bumps any non-exact quotient up to the next integer before the floor chops it down — quietly turning a round-down into a round-up.

1.6 Remark — Ceiling Is the “Containers Needed” Function

This problem is the prototype of a pattern that appears constantly: *how many fixed-capacity containers are needed to hold a given amount?* The answer is always a ceiling. For n units of stuff and bins of capacity c ,

$$\text{bins needed} = \left\lceil \frac{n}{c} \right\rceil.$$

Sand into bags, students onto buses, bytes into disk blocks, pages of items at a fixed-size-per-page — all the same. The ceiling captures the universal truth that a *partly used* container still counts as a whole container. (Floor, by contrast, answers the different question “how many *completely full* containers do I get?” — here, **6**.)

1.7 Remark — It’s Division With Remainder

The ceiling is doing nothing more than the quotient-remainder split. Dividing **340** by **50**:

$$340 = 6 \cdot 50 + 40, \quad \text{quotient } 6, \text{ remainder } 40.$$

The quotient **6** counts the full bags; the *nonzero* remainder **40** means there is leftover sand, forcing one more bag. In general,

$$\left\lceil \frac{n}{c} \right\rceil = \begin{cases} \lfloor n/c \rfloor & \text{if } c \mid n \text{ (remainder } 0), \\ \lfloor n/c \rfloor + 1 & \text{if } c \nmid n \text{ (remainder } > 0). \end{cases}$$

Here the remainder is $40 > 0$, so we land in the second case: $6 + 1 = 7$. Had the load been exactly **300** pounds (remainder **0**), floor and ceiling would coincide at **6** — no “extra” bag, and $\lceil 300/50 \rceil = \lfloor 300/50 \rfloor = 6$.

1.8 Remark — Floor vs Ceiling: Which Question Are You Asking?

Floor and ceiling differ by *at most* 1 (they agree exactly when the quotient is already a whole number), but choosing the wrong one is a classic off-by-one error. The rule of thumb:

You want...	Use	Here
how many bags are <i>needed</i> to hold it all	ceiling	$\lceil 340/50 \rceil = 7$
how many bags can be <i>completely filled</i>	floor	$\lfloor 340/50 \rfloor = 6$

Table 1.1

The phrase “bags **required**” in the problem is the tell: requiring all the sand to be bagged means no leftover is allowed, which is the round-*up*, ceiling, reading. If the question had instead asked “how many bags can be filled to capacity,” the answer would be the floor, **6**.